

ArgiSense pH Firmware Overview

This document describes the pH product profile on the `argisense_ph_u575rg` board. It is intentionally separate from `docs/methane-pressure-firmware-overview.md` so pH-specific analog-front-end decisions do not get mixed with the Dymament methane sensor flow.

Board Target

`argisense_ph_u575rg`

The board is based on STM32U575RGT6 and the Sensor Platform V1 `pH_V1_0` schematic. It keeps the same boot, flash, clock, debug, RS485, storage, and service concepts as the methane + pressure board, but maps the measurement front-end to pH-specific ADC/DAC hardware.

Selected MCU Mapping

The schematic exports the pH analog-front-end nets as sheet-level ports (`ADC_SPI_*`, `ADC_RESET`, `ADC_START`, `ADC_SRDY`, `DAC_I2C_*`, `DAC_RESET`). The firmware mapping below assigns those ports to STM32U575RGT6 pins while keeping USB-C, SWD, debug UART, and RS485 away from pH analog traffic:

Function	Schematic signal	STM32U575RGT6 mapping	Zephyr node/property
Debug console	STLK_UART_TX/RX	PA9/PA10	<code>usart1</code>
RS485 TX/RX/DE	O_C_UC_UART2_TXD, I_C_UC_UART2_RXD, O_C_UC_RS485_RnD1	PA2/PA3/PA1	<code>usart2</code> , <code>rs485_modbus</code>
RS485 termination	RS485_RESISTOR_EN	PB0	<code>rs485-termination-gpios</code>
USB-C device	USB2_P, USB2_N	PA12, PA11	<code>otgfs_snippets</code>
SWD	MCU_SWDI0, MCU_SWCLK	PA13, PA14	<code>debugger</code>
pH ADC SPI CS/SCLK/DOUT/DIN	ADC_SPI_CS/SCLK/DOUT/DIN	PA4/PA5/PA6/PA7	<code>ph_ads124s08</code> on <code>spi1</code>
pH ADC reset/start/ready	ADC_RESET, ADC_START, ADC_SRDY	PB12, PB13, PB15	<code>reset-gpios</code> , <code>start-sync-gpios</code> , <code>drdy-gpios</code>
pH bias DAC I2C	DAC_I2C_SCL, DAC_I2C_SDA	PC0, PC1	<code>ph_ad5675</code> on <code>i2c3</code>
pH bias DAC reset	DAC_RESET	PC2	<code>reset-gpios</code>
pH bias DAC LDAC	LDAC*	Not routed to MCU on <code>pH_V1_0</code>	R73 0R ties LDAC low; optional <code>ldac-gpios</code> for future PCB
ADC clean 3V3 enable	U_O_D_3V3_EN	PC3	<code>adc-power-gpios</code>
Electrode zero switch	ELECTRODE_ZERO_CTRL	PC4	<code>electrode-zero-gpios</code>
MCU analog monitor	MCU_ADC3_INP0 equivalent	PC5 / ADC1_IN14	<code>mcu-adc</code>
GP8302 DAC0	O_D_UC_I2C1_SCL/SDA	PB8/PB9	<code>current-loop-dac0</code>
AT24C512C EEPROM	MCU_I2C1_SCL/SDA	PB8/PB9	<code>eeprom0</code>
GP8302 DAC1	O_D_UC_I2C2_SCL/SDA	PB10/PB14	<code>current-loop-dac1</code>
Analog rail enable	U_O_D_ANA_PWR_EN	PC9	<code>analog-power-gpios</code>
DAC boost enable	O_D_UC_PWR_DAC	PC6	<code>dac-power-gpios</code>
DAC alarms	O_D_DAC_ALARM1/2	PC7, PB2	<code>dac-alarm-gpios</code> , <code>dac1-alarm-gpios</code>
Sensor/pre rail enable	+3V3_PRE control	PC8	<code>pre-power-gpios</code>

The pH board uses the same `storage_partition` as the methane + pressure board for Zephyr Settings/NVS through `zephyr, settings-partition`. For the consolidated IO mapping and conflict list across both product variants, see `docs/sensor-io-mapping.md`.

pH Measurement Hardware

Device	Bus	Address/CS	Purpose
ADS124S08IPBSR	spi1	CS0 / PA4	24-bit ADC for ISFET, REFET, PT1000, and gate monitor
AD5675BCPZ-REEL7	i2c3	0x0c	8-channel pH bias DAC
REF3025AIDBZR	local reference	n/a	2.5 V reference
LT3582EUD-5	i2c2 placeholder	0x2c placeholder	+5V_AN and -5V_AN; node remains disabled until address/programming is verified
GP8302 channel 0	i2c1	0x58	4-20 mA output for pH
GP8302 channel 1	i2c2	0x58	4-20 mA output for temperature or secondary value
AT24C512C	i2c1	0x50	External EEPROM
RS485 Modbus RTU	usart2	n/a	Service data, configuration, and future pH DFU/register map

Schematic notes checked for the pH front-end:

- ADS124S08 is on sheet 10 and uses SPI plus RESET, START/SYNC, and DRDY.
- PT1000 is routed to ADS124S08 AIN2/AIN3, with a schematic note for 250 uA excitation current on AIN2.
- The pH product does not populate or use an HTU21D humidity IC. Ambient/process temperature for pH compensation comes from the PT1000 path through ADS124S08.
- AD5675 is on sheet 11, I2C address 0x0c, with an external REF3025 2.5 V reference. Firmware configures AD5675 gain x1, so the code-to-output span is 0 . . 2 . 5 V.
- AD5675 LDAC* is not MCU-controlled on the current schematic. It is pulled low by the R73 0 ohm option and exposed at TP17. The driver supports optional ldac-gpios, but the current board DTS intentionally omits it until a PCB revision routes LDAC* to an MCU pin.
- AD5675 VOUT0/VOUT2 nominal setpoints are 448 mV for ISFET/REFET current sources; VOUT1/VOUT3 nominal setpoints are 500 mV for ISFET/REFET VDS control.

ADS124S08 Channel Labels

Label	Role
AIN0_Vs_ISFET	ISFET source measurement
AIN1_Vs_REFET	REFET source measurement
AIN2_PT1000_P	PT1000 positive input
AIN3_PT1000_N	PT1000 negative input
ISFET_BULK_ADC	ISFET bulk monitor
ISFET_SUBS_ADC	ISFET substrate monitor
REFET_BULK_ADC	REFET bulk monitor
REFET_SUBS_ADC	REFET substrate monitor
ISFET_DRAIN_ADC	ISFET drain monitor
REFET_DRAIN_ADC	REFET drain monitor
GATE_BUFF_ADC	Gate buffer output

PT1000 uses the ADS124S08 IDAC path with 250 uA.

AD5675 Bias Outputs

Output	Role
VOUT0_ID_SET_ISFET	ISFET current source setpoint, nominal 448 mV
VOUT1_VDS_SET_ISFET	ISFET VDS setpoint, nominal 500 mV
VOUT2_ID_SET_REFET	REFET current source setpoint, nominal 448 mV
VOUT3_VDS_SET_REFET	REFET VDS setpoint, nominal 500 mV
VOUT4_BULK_ISFET	ISFET bulk bias, tracks Vs_ISFET reading
VOUT5_SUBS_ISFET	ISFET substrate bias, tracks Vs_ISFET reading
VOUT6_BULK_REFET	REFET bulk bias, tracks Vs_REFET reading
VOUT7_SUBS_REFET	REFET substrate bias, tracks Vs_REFET reading

pH Calculation Formula and Influencing Parameters

The firmware publishes `ph_x1000`, where a value of 7000 means pH 7.000. The pH result is intentionally invalid until calibration mode is selected and `ph_calibration_valid` is set. This prevents raw ADC voltage from being mistaken for a calibrated chemical measurement.

Raw pH Signal

The ADS124S08 measurement layer reads source and gate voltages first:

```
VGS_ISFET_uV = gate_uV - vs_isfet_uV
VGS_REFET_uV = gate_uV - vs_refet_uV
```

The selected calibration mode then decides which raw signal feeds the pH equation:

```
mode 1, ISFET-REFET differential:
    ph_raw_uv = VGS_ISFET_uV - VGS_REFET_uV
               = vs_refet_uV - vs_isfet_uV

mode 2, reference-electrode:
    ph_raw_uv = VGS_ISFET_uV
```

For the current ISFET/REFET flow, mode 1 is the preferred bring-up path because the REFET channel helps cancel common-mode drift from the shared gate/reference node. Mode 2 is kept for a future reference-electrode topology or lab diagnostic mode.

PT1000 Temperature Compensation

PT1000 temperature is measured by ADS124S08 using the configured IDAC current from devicetree. The current pH board uses 250 uA.

```
R_pt1000_mohm = pt1000_uv * 1000 / pt1000_idac_ua
temperature_centi_c = (R_pt1000_mohm - 1000000) * 100 / 3850
```

This is a linear IEC 60751 bring-up approximation around 0 degC. It is suitable for firmware integration and early hardware bring-up, but final production calibration should validate the PT1000 curve over the expected process temperature range.

PT1000 Bring-up and Service Reads

The pH firmware reads the PT1000 as a differential ADS124S08 measurement:

```
ADS124S08 positive input: AIN2_PT1000_P
ADS124S08 negative input: AIN3_PT1000_N
ADS124S08 IDAC output:    AIN2_PT1000_P
IDAC current:             250 uA
```

The direct shell command for this path is:

```
argisense ph adc 2 3 2
```

This means: read AIN2 - AIN3 and enable the ADS124S08 IDAC on AIN2. The normal measurement cycle uses the same path, then publishes the converted temperature and raw PT1000 voltage into the pH service register map.

Useful service checks:

```
argisense ph sample
argisense sensors
argisense rs485 12 2
argisense rs485 71 1
argisense rs485 104 3
```

Register meaning:

Register	Meaning
12..13	temperature_centi_c, signed 32-bit PT1000 temperature
71	temperature_last_error, signed 16-bit last PT1000 read/conversion error
75	ph_adc_status; bit 2 is set when PT1000 conversion succeeded
104..105	pt1000_uv, signed 32-bit measured PT1000 voltage
106	ph_operating_status; includes pH operating-point/calculation bits

Expected PT1000 voltage with 250 uA excitation:

Temperature	Approx. PT1000 resistance	Approx. differential voltage
0 degC	1000 ohm	250 mV
25 degC	1096 ohm	274 mV
80 degC	1308 ohm	327 mV

During bring-up, a reported temperature around -259.74 degC usually means pt1000_uv is 0, so the firmware is seeing no differential PT1000 voltage. Check that PT1000 is actually connected between ADS124S08 AIN2/AIN3, the IDAC path can source current into the PT1000 network, ADC_RESET, ADC_START, ADC_SRDY, SPI, the ADC power rail, and the REF3025 reference are all valid.

The pH slope is adjusted by temperature before the final pH calculation:

```
delta_centi_c = temperature_centi_c - ph_temp_reference_centi_c
temp_multiplier_ppm =
    1000000 + ph_temp_coeff_ppm_per_c * delta_centi_c / 100

slope_temp_uv_per_ph =
    ph_slope_uv_per_ph * temp_multiplier_ppm / 1000000
```

If PT1000 temperature is not valid, firmware uses ph_temp_reference_centi_c instead and the TEMP_COMP_USED operating-status bit is not set.

Final pH Equation

The calibrated pH equation used by firmware is:

```
ph_base_x1000 =
    7000 + (ph_raw_uv - ph7_raw_uv) * 1000 / slope_temp_uv_per_ph

ph_x1000 =
    ph_base_x1000 * ph_slope_x1000 / 1000 + ph_offset_x1000

pH = ph_x1000 / 1000.0
```

ph7_raw_uv is the raw voltage measured at pH 7. ph_slope_uv_per_ph is the raw voltage change per pH unit after calibration. Its sign must match the real sensor response: if pH increases while ph_raw_uv decreases, store a negative slope. The default 59000 uV/pH is only a bring-up placeholder and the pH value remains invalid until calibration is enabled.

Validity Gates

pH output is marked valid only when all of these conditions pass:

- ph_calibration_mode is not disabled.
- ph_calibration_valid is 1.
- abs(slope_temp_uv_per_ph) is at least 1000 uV/pH.
- Raw ISFET/REFET reads are available.
- In ISFET-REFET mode, both ISFET and REFET VDS checks pass.
- In reference-electrode mode, the ISFET VDS check passes.

The operating-point check uses:

```
VDS_ISFET_uv = isfet_drain_uv - vs_isfet_uv
VDS_REFET_uv = refet_drain_uv - vs_refet_uv
target VDS   = -500000 uV
tolerance    = +/-200000 uV
```

If these checks fail, the firmware keeps the pH value invalid and the current loop uses the configured fault current instead of a misleading process value.

Parameters That Affect pH

Parameter	Default / source	Effect
ph_calibration_mode	default 0, disabled	Selects disabled, ISFET-REFET differential, or reference-electrode raw signal.
ph_calibration_valid	default 0	Enables calibrated pH publishing after calibration data is known good.
ph7_raw_uv	default 0 uV	Raw voltage offset captured in pH 7 buffer.
ph_slope_uv_per_ph	default 59000 uV/pH	Main raw slope. Set sign and magnitude from two-point or multi-point pH calibration.
ph_temp_reference_centi_c	default 2500	Reference temperature for the calibrated raw slope, normally 25.00 degC.
ph_temp_coeff_ppm_per_c	default 3354 ppm/C	Temperature coefficient applied to the raw slope before pH calculation.
ph_slope_x1000	default 1000	Final gain trim after the raw pH conversion.
ph_offset_x1000	default 0	Final pH offset trim after the raw pH conversion.
temperature_centi_c	PT1000 through ADS124S08	Enables temperature compensation when valid.
vs_isfet_uv, vs_refet_uv, gate_uv	ADS124S08 reads	Produce VGS and the raw pH signal.
isfet_drain_uv, refet_drain_uv	ADS124S08 reads	Used to verify VDS operating point before pH is accepted.
VOUT4..VOUT7	AD5675 source tracking	Bias bulk/substrate outputs from the measured source voltages, affecting analog operating point.
ph_range_low_x1000, ph_range_high_x1000	default 0..14000	Do not change the pH calculation; only map calibrated pH to DAC0 4-20 mA output.

Suggested Calibration Sequence

1. Build and flash with the pH firmware profile.
2. Use `argisense_ph_sample` and raw RS485 registers to confirm the ADC, AD5675 source tracking, PT1000 temperature, and VDS status bits are stable.
3. Put the probe in a pH 7 buffer, wait for settling, then store the measured `ph_raw_uv` as `ph7_raw`.
4. Put the probe in a second buffer, for example pH 4 or pH 10, and compute:

$$\text{ph_slope_uv_per_ph} = (\text{ph_raw_uv_buffer2} - \text{ph7_raw_uv}) / (\text{ph_buffer2} - 7.0)$$

5. Set `ph_cal_mode` to 1 for ISFET-REFET differential mode, write the calibrated `ph7_raw` and `ph_slope_uv`, then set `ph_cal_valid` to 1.
6. Recheck pH 7 and the second buffer. Use `ph_slope` and `ph_offset` only for small final production trims after the raw slope and pH7 offset are correct.

Current Firmware State

The board now builds with a dedicated pH firmware profile under `app/products/ph/src`. It verifies the pH devicetree topology, initializes persistent pH settings, starts a Modbus RTU server on RS485, publishes a pH-specific holding-register map, and registers pH shell diagnostics.

The pH firmware uses the same service connectors as the methane + pressure firmware:

- USART2 RS485 for Modbus configuration, live-data reads, service commands, and RS485 DFU register writes.
- USB-C CDC ACM for shell/console with the `usbconsole` profile.
- USB-C composite CDC ACM for shell/console plus MCUMgr firmware update with the `usbupdate` profile.

ADS124S08 and AD5675 protocol drivers are now implemented as out-of-tree Zephyr module drivers:

- `drivers/adcs/ads124s08.c` provides reset, register read/write, single-shot conversion, DRDY polling, raw 24-bit sign extension, raw-to-microvolt conversion, and a minimal Zephyr ADC API surface.
- `drivers/dacs/ad5675_i2c.c` exposes the AD5675 as an 8-channel 16-bit Zephyr DAC API device and writes each channel with the AD5675 write-and-update I2C command.
- `app/products/ph/src/ph_measurement.c` powers the pH analog rails, applies safe initial AD5675 bias values, reads ISFET/REFET source and gate, updates VOUT4..VOUT7 from the measured source voltages, reads drain, bulk, substrate, and PT1000 diagnostics, then publishes raw voltages, temperature, pH status, and calibration-gated pH into the pH register snapshot.
- `app/products/ph/src/current_loop_output.c` maps the measured values to the two GP8302 current-loop outputs: DAC0 outputs calibrated pH and DAC1 outputs PT1000 temperature.

The pH value remains invalid until calibration is explicitly enabled. The firmware stores pH7 raw offset, raw slope in $\mu\text{V/pH}$, temperature reference, temperature coefficient, calibration mode, and calibration-valid state in NVS. This avoids publishing a pH value that looks valid but is only a raw voltage placeholder.

DAC output behavior:

Output	Source	Valid behavior	Invalid behavior
GP8302 DAC0	Calibrated pH, ph_x1000	Maps ph_range_low_x1000..ph_range_high_x1000 to dac_min_ua..dac_max_ua	Outputs dac_fault_ua until pH calibration is valid
GP8302 DAC1	PT1000 temperature, temperature_centi_c	Maps temperature_range_low_centi_c..temperature_range_high_centi_c to dac_min_ua..dac_max_ua	Outputs dac_fault_ua if PT1000 read fails

With defaults, DAC1 maps 0.00..80.00 degC to 4..20 mA. Change this range from shell or RS485 with the temp_low and temp_high settings.

Build command:

```
py -3.12 -m west build -p always -b argisense_ph_u575rg argisense-zephyr-app
```

Equivalent project script command:

```
argisense-zephyr-app\app\compile.bat ph
```

USB-C shell and update builds:

```
argisense-zephyr-app\app\compile.bat ph usbconsole
argisense-zephyr-app\app\compile.bat mcuboot ph usbupdate
argisense-zephyr-app\app\compile.bat mcuboot ph hsi usbupdate
```

System Illustrations

pH Firmware Architecture

ADS124S08 acquisition, AD5675 biasing, shared RS485/USB services

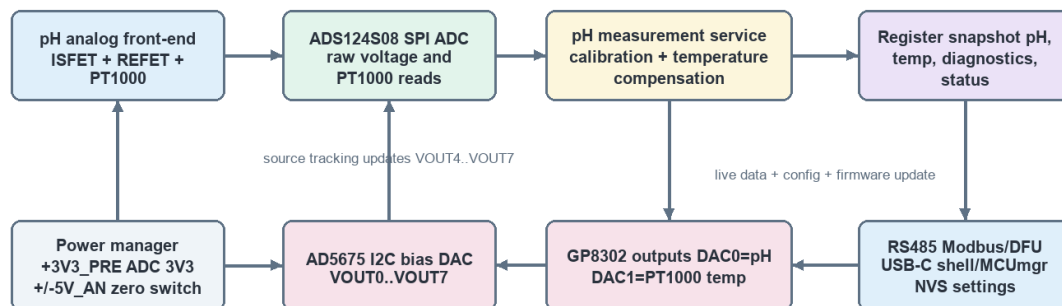


Figure 1. pH firmware architecture and shared service blocks.

pH firmware architecture with ADS124S08 acquisition, AD5675 biasing, RS485/USB service, settings, and dual GP8302 outputs.

pH Measurement Cycle

Rail sequencing, ADS124S08 sampling, AD5675 source tracking, and GP8302 outputs

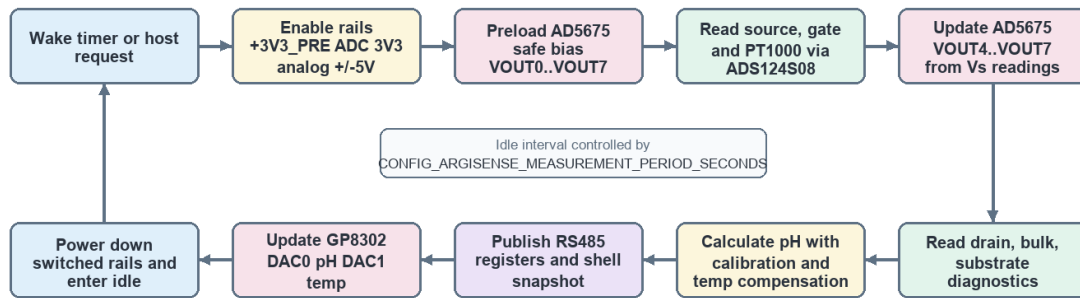


Figure 2. pH measurement, bias tracking, output update, and low-power idle loop.

pH measurement cycle showing rail enable, AD5675 bias preload, ADS124S08 reads, source-tracking bias updates, pH calculation, output update, and idle.

pH Output and Communication Mapping

Calibrated pH, PT1000 temperature, diagnostics, RS485, USB-C, and current loops

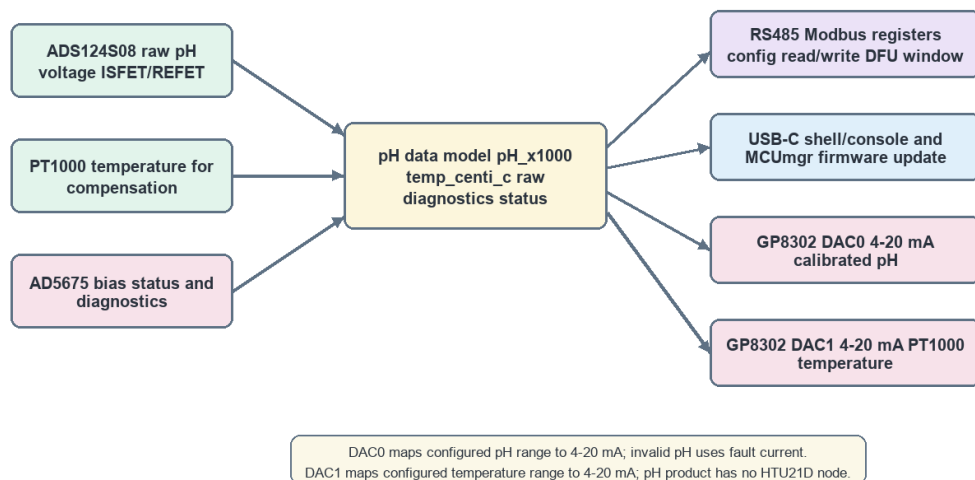


Figure 3. pH values, diagnostics, service interfaces, and dual current-loop outputs.

pH output and communication mapping showing calibrated pH, PT1000 temperature, raw diagnostics, RS485/USB service paths, and dual 4-20 mA outputs.

pH Service Register Map

The pH firmware uses device ID 0xA652 and register map version 3. Multi-word values are high word first.

Range	Purpose
0 . . 38	Identity, status, live pH/temperature snapshot, firmware version, RS485 serial settings, calibration mode, calibration-valid flag, and service command register
40 . . 67	pH range, temperature range, output correction slope/offset, raw pH7 offset, raw uV/pH slope, temperature compensation settings, and DAC trim settings
70 . . 76	pH and PT1000 temperature error/status registers; 72 . . 74 are reserved because the pH product has no HTU21D humidity sensor
80 . . 106	Raw pH diagnostics: raw pH voltage, VGS ISFET/REFET, source/gate/drain/bulk/substrate voltages, PT1000 voltage, and operating-point status
1000 . .	Shared RS485 DFU window when built with MCUboot and CONFIG_ARGISENSE_RS485_DFU=y

Writable RS485 settings are stored through Zephyr Settings/NVS under the pH namespace `argisense_ph/config`, separate from the methane + pressure settings namespace. Baudrate, unit ID, data-bit, parity, and stop-bit writes are saved immediately but require reboot before the running Modbus server uses the new transport parameters.

Shell commands:

```
argisense drivers
argisense sensors
argisense rs485
argisense rs485 1000 36
argisense settings
argisense settings get ph_high
argisense settings set ph_cal_mode 1
argisense settings set ph7_raw 0
argisense settings set ph_slope_uv 59000
argisense settings set ph_cal_valid 1
argisense settings set temp_low 0
argisense settings set temp_high 8000
argisense settings set address 2
argisense settings reset
argisense ph adc 0
argisense ph adc 2 3 2
argisense ph dac 0 448
argisense ph zero 1
argisense ph sample
```

Proposed pH Measurement Flow

1. Power and settle:

enable +3V3_PRE, ADC clean 3.3 V, and +/- 5V_AN; wait the configured analog settle time.

2. Safe bias preload:

configure all eight AD5675 channels, then set VOUT0/VOUT2 to 448 mV and VOUT1/VOUT3 to 500 mV. Preload VOUT4..VOUT7 to 0 mV before the first ADC source reading.

3. Electrode zero check:

use `ELECTRODE_ZERO_CTRL` as a service/calibration mode to short or release the electrode input path, then capture ADC offset data.

4. Temperature read:

configure ADS124S08 for AIN2/AIN3 differential, enable 250 uA IDAC on AIN2, read the 24-bit conversion, convert voltage to PT1000 resistance, then use the PT1000 temperature equation. The current firmware uses a linear bring-up approximation around 0 degC.

5. ISFET/REFET read:

read `AIN0_Vs_ISFET`, `AIN1_Vs_REFET`, and `GATE_BUFF_ADC`, then rewrite VOUT4/VOUT5 to the measured `Vs_ISFET` and VOUT6/VOUT7 to the measured `Vs_REFET`. After the source-tracking outputs settle, read drain, bulk, and substrate monitor channels. The current raw differential value is `VGS_ISFET - VGS_REFET`, which simplifies to

$V_{s_REFET} - V_{s_ISFET}$ when both FETs share the same gate/reference node.

6. Closed-loop pH calculation:

verify the ISFET/REFET VDS monitor is near -0.5 V , then calculate pH only when calibration mode and calibration-valid are enabled. Calibration stores pH7 raw offset, $\mu\text{V/pH}$ slope, and temperature compensation settings.

7. Publish and output:

write the final calibrated pH and PT1000 temperature into the RS485 register map, update GUI graph values, then map pH to GP8302 DAC0 and PT1000 temperature to GP8302 DAC1.

Next Firmware Work

1. Validate ADS124S08 SPI mode, DRDY timing, REF3025 reference polarity, and PT1000 IDAC routing on hardware.
2. Calibrate real pH7 raw offset and slope using buffer solutions and confirm the sign of `ph_slope_uv_per_ph`.
3. Validate the VDS tolerance window with a DMM/scope before relying on the `ph_valid` bit.
4. Add optional GUI diagnostic fields for gate voltage, ISFET/REFET voltages, PT1000 raw voltage, and pH output current after the raw diagnostic UX is finalized.